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Application Note

2013/04/09

Product : Vortex86EX (Vx86EX-9123)

Subject : Compatibility with eMMC on Vx86EX-9123.

Description:
When using eMMC with Vortex86EX, the IDE status error will happen randomly.

Root Cause:
The eMMC design on Vortex86EX is followed SD version 2.0 specification which the define of Data Response Time (NAC) is less than 100ms. Nowadays in eMMC specification version 4.4, the Data Response Time (NAC) is defined by following formula, and it might over 100ms.

N_{AC}	2	$10 * (TAAC * F_{OP} + 100 * NSAC)^1$	clock cycles

Solutions:
- Use DMP approved eMMC from Greenliant (http://www.greenliant.com/), the part number of Greenliant is *G85VM0512P-S-I-LFWE-PP038* which is a 512MB SLC eMMC with custom firmware that Data Response Time (NAC) is less than 100ms.

- Make sure with your eMMC vendor that the Data Response Time (NAC) is less than 100ms or they can modify the firmware to fit it.

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